

GNN Final Report

Target: BACC > 0.94

Phase 1: Optimization (5-Fold CV)

Best Average CV Score: 0.9552

(This score determined the best parameters below)

Best Hyperparameters Found:

model_type: GAT
num_layers: 6
hidden_channels: 256
heads: 2
dropout: 0.21
lr: 2.87e-04
batch_size: 32
weight_decay: 2.62e-04

Phase 2: External Validation (Final Check)

Tested on 'moleculas_unicas.csv' (never seen during training)

External BACC (Thr 0.5): 0.8916
External BACC (Optimal): 0.9124 (Thr=0.94)
Target Met (>0.94): NO ☐

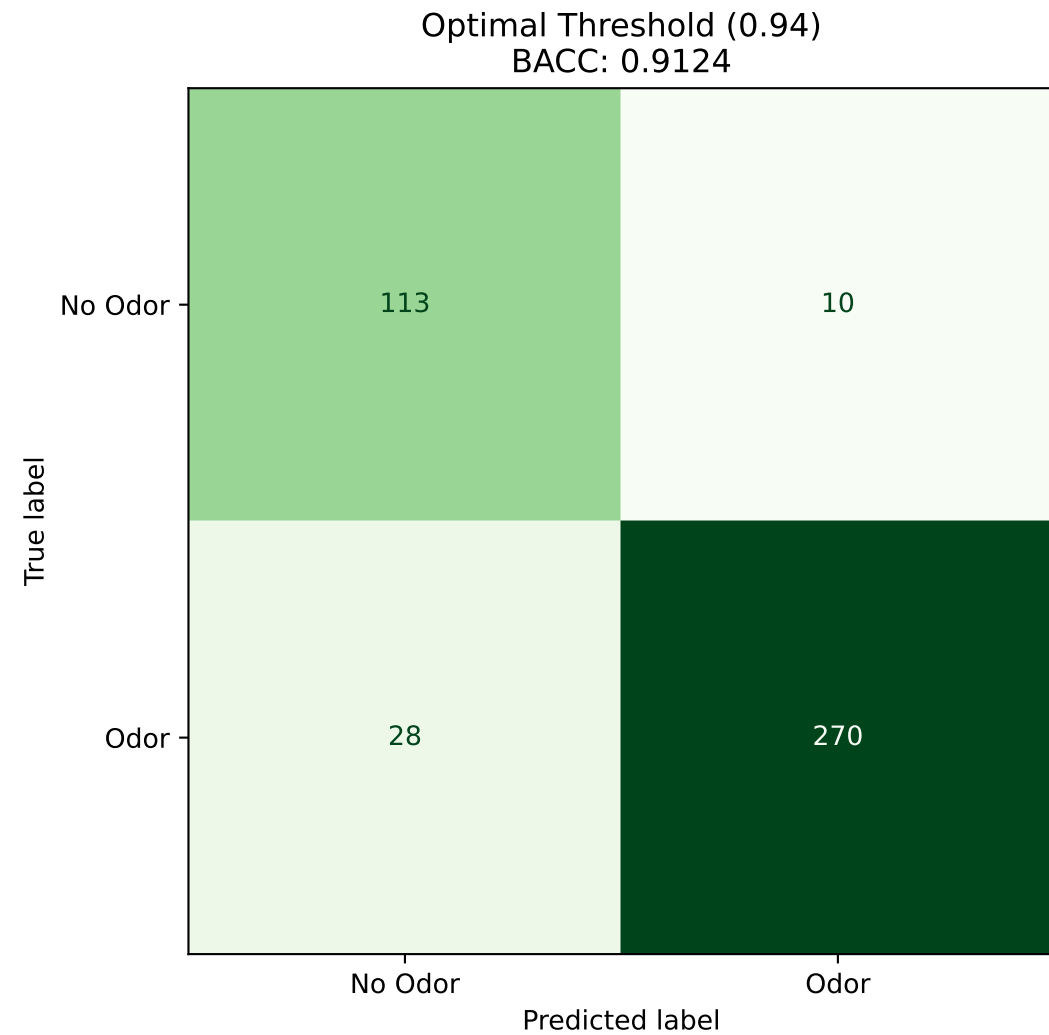
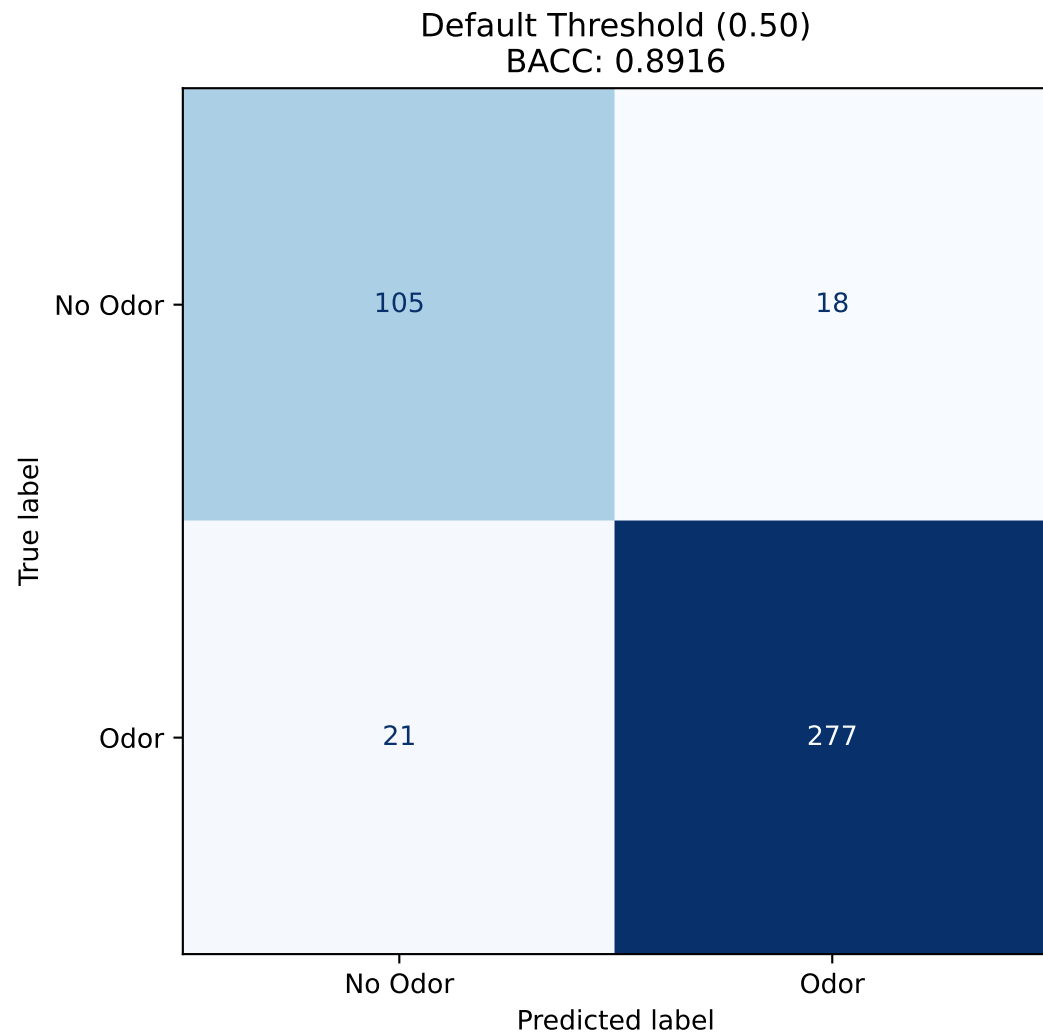
Detailed External Metrics:

[Threshold 0.5]	[Optimal Thr 0.94]
MCC: 0.78	MCC: 0.79
Precision: 0.94	Precision: 0.96
Recall: 0.93	Recall: 0.91
F1 Score: 0.93	F1 Score: 0.93

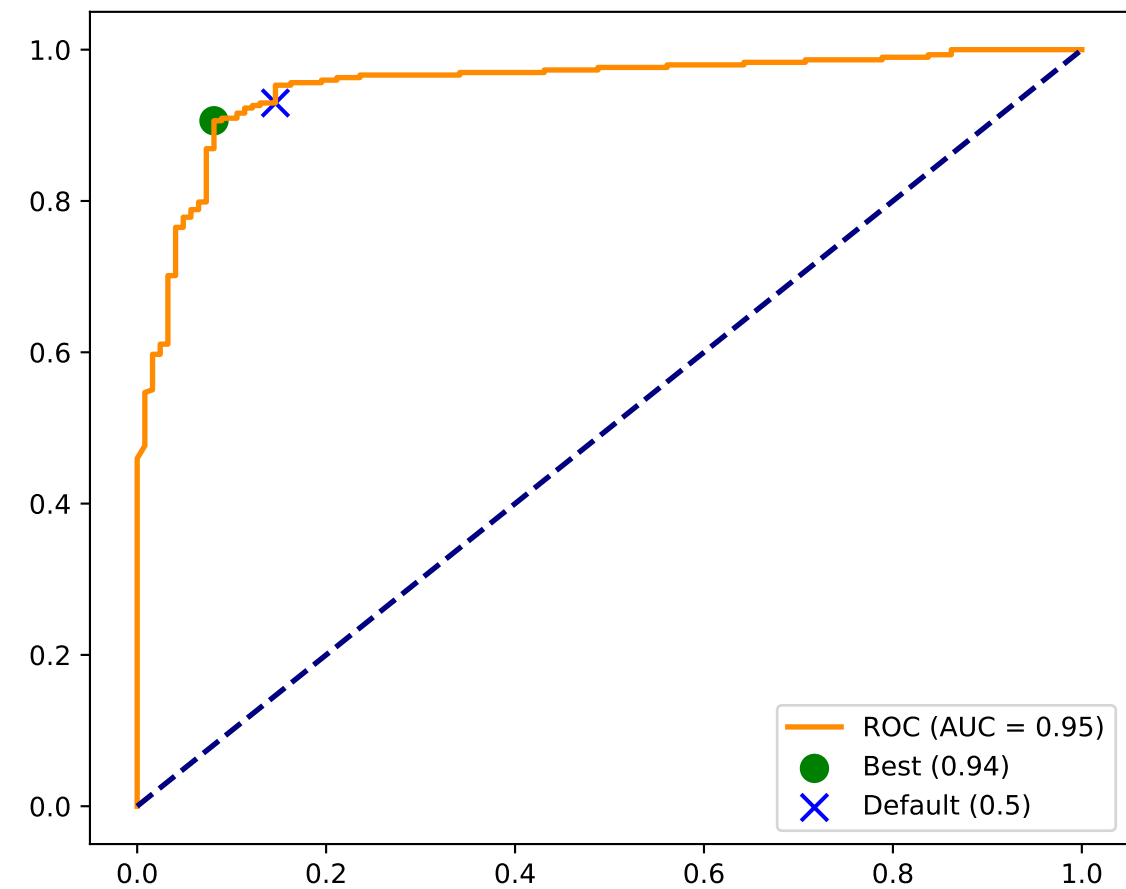
Dataset Statistics:

Train Set: 3791 (Pos:3240 Neg:551)
Ext Set: 421 (Pos:298 Neg:123)

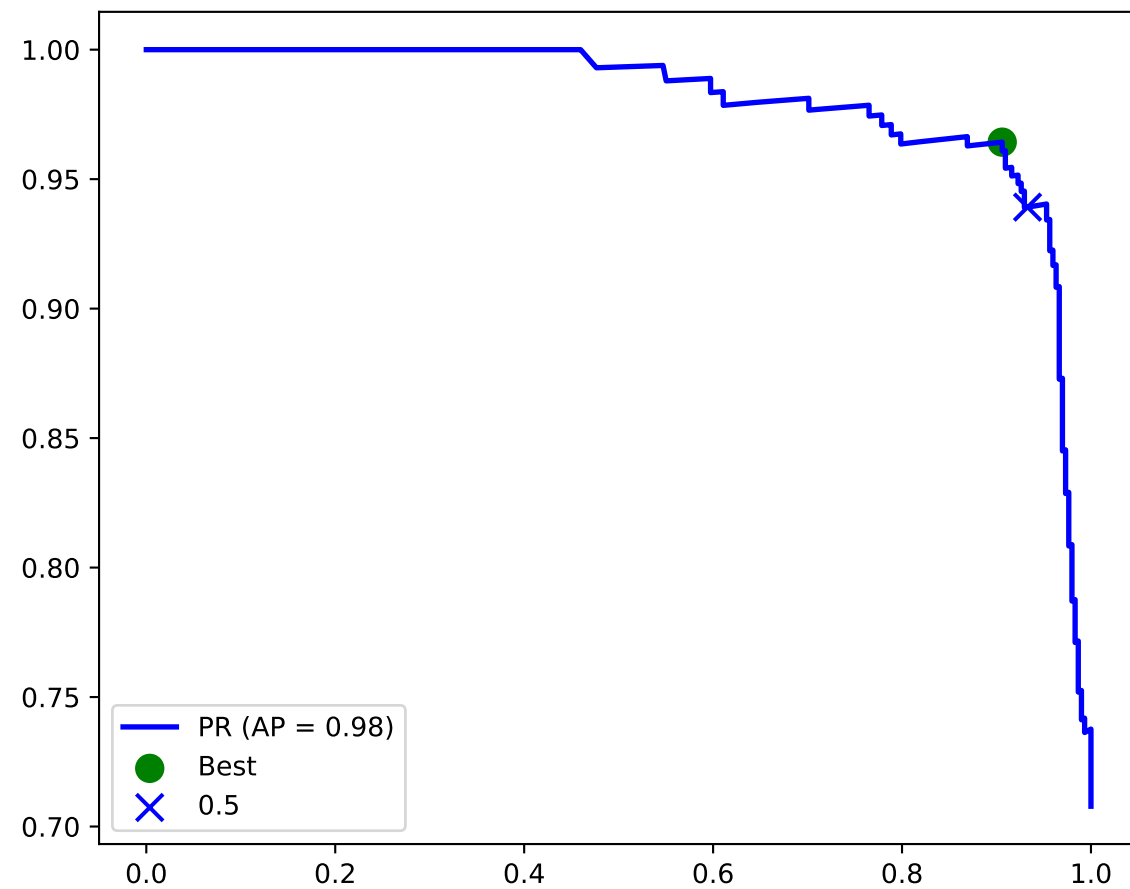
Confusion Matrices Comparison



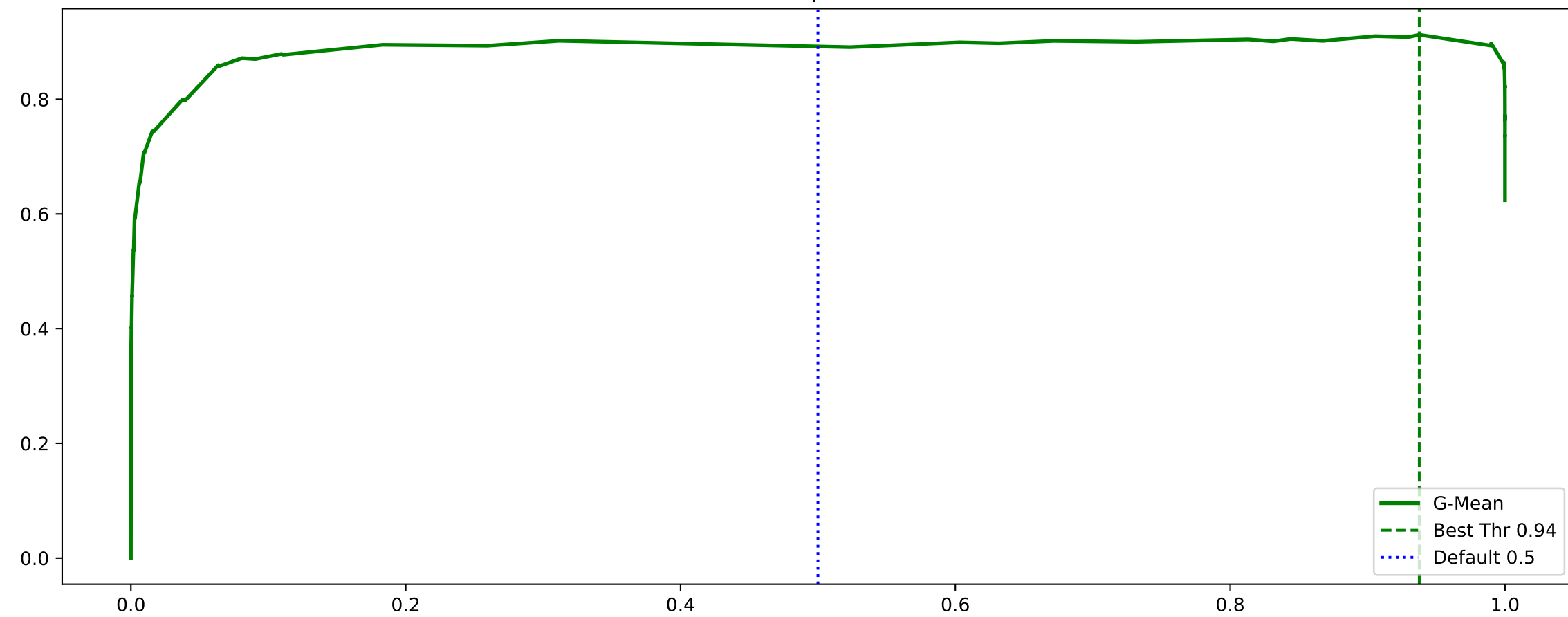
ROC Curve



Precision-Recall Curve



G-Mean Optimization



GNN Architecture Diagram

